

Alexander Berger-Liedtka

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Overview

Experienced software engineer with a strong track record building ML-enabled systems for real-world video and sensor data in research-driven environments.

Over a decade of experience taking ML systems from research prototype to production-grade, with deep **hands-on work in PyTorch model training, evaluation, and deployment.**

Comfortable **owning problems end-to-end**, from model experimentation and training pipelines through deployment, operations, and long-term maintainability.

Recent Experience

Senior Scientific Software Engineer, The Jackson Laboratory – Bar Harbor, ME 2025 – present

- **The Kumar Lab Role** (May 2025 – present)

- Improved a machine learning processing workflow from a theoretical ~2 days to ~8 hours end-to-end.
- Scaled mixed CPU/GPU execution (A100 MIG slicing with V100 fallback), supporting ~100 concurrent GPU jobs and thousands of CPU jobs with *improved stability* under OOM and user-error failure modes.
- Designed, trained, and productionized PyTorch keypoint models (arena corners; lixit/drinker), integrated into the pipeline with MLflow-tracked experimentation.
- Operationalized a dedicated workflow to re-encode and compress ~2.1 PB of legacy raw video into a ~55 GB long-term retention format, balancing quality and storage cost and enabling future processing and re-analysis.
- Developed evaluation methodology and tooling for model comparison across video datasets, enabling systematic experimentation and reproducible results.
- Built data infrastructure for managing and processing large-scale video corpora (hundreds of thousands of hours), including staging, validation, and format standardization.

- **Software & Platforms Architecture Role** (Nov 2025 – present)

- Selected to lead systems architecture across the Software & Platforms department, defining technical direction for shared infrastructure spanning ML workflows, data processing, and cloud/HPC compute.
- Designed integrated data science infrastructure connecting experiment tracking, model training, and large-scale data pipelines across heterogeneous compute environments.

Technical Leadership

Owned the technical direction for a laboratory mouse-behavior video ML pipeline on HPC (Nextflow+Slurm), driving it from stalled prior attempts to reliable production execution. Defined the execution architecture, resource strategy, and failure-handling approach that enabled largely unattended runs.

Led the end-to-end model integration strategy for the behavioral pipeline, making decisions on training infrastructure, model selection criteria, and the MLflow experimentation workflow that governed how models were evaluated and promoted to production.

Designed the evaluation framework and methodology used for systematic model comparison across video datasets, establishing the benchmarking standards the team uses to validate keypoint detection and inference performance.

Defined the technical approach for scalable video data pipelines processing hundreds of thousands of hours, including the architecture for format standardization, quality validation, and the large-scale re-encoding workflow for legacy data.

Core Competencies

ML Systems & Experimentation: PyTorch, MLflow, evaluation metrics, model-to-pipeline integration, training optimization, data/compute optimization

Workflow Orchestration & HPC: Nextflow, Slurm, Bash, resource profiles, retries/escalation, data staging, high-throughput batch processing, A100/A100-MIG/V100

Reliability & Operations: Failure-mode analysis, OOM mitigation, guardrails to reduce user error, unattended execution, incident-driven improvements

Distributed Systems & APIs: Python, Rust, Typescript, FastAPI, SQLAlchemy, Pydantic, Postgres/PgVector, Redis, async job orchestration patterns, durable orchestration

Cloud & Infrastructure: Kubernetes, GCP, Docker, Terraform, Helm/Skaffold, CI/CD, Prometheus/Grafana

Leadership & Strategy: Technical strategy, technical leadership, product management, cross-functional execution, mentorship, governance/standards, architectural reviews and stakeholder alignment

Additional Experience

Senior Scientific Software Engineer, The Jackson Laboratory – Bar Harbor, ME Aug 2022 – 2025

- Led design and deployment of event-driven and service-oriented architectures on Kubernetes with Terraform-managed infrastructure, enabling 13x increase in system release cadence.
- Mentored engineers and drove cloud-native standards and design to improve velocity and system reliability.
- Expanded technical leadership role across platforms, shaping departmental initiatives.

Scientific Software Engineer, The Jackson Laboratory – Bar Harbor, ME Sept 2018 – Aug 2022

- Built reusable genomics services, APIs, and data tooling; improved platform consistency through testing, versioning, and workflow standards.

Associate Scientific Software Engineer, The Jackson Laboratory – Bar Harbor, ME Jan 2016 – Sept 2018

- Delivered production components for scientific workflows and data products; contributed to design reviews, performance improvements, code quality initiatives, and documentation.

Software Engineer, Prior Software Engineering Experience June 2013 – Jan 2016

Selected Projects

Behavioral Video ML Pipeline

- Production ML pipeline for laboratory mouse-behavior video data (1-hour videos, 30 FPS, 800x800) executed on Slurm/HPC with Nextflow and mixed CPU/GPU stages.
- Improved keypoint inference characteristics (e.g., lixit/drinker keypoint model: PCK@0.01 = 0.9974; ~545 ms/frame CPU inference), increasing robustness of downstream pipeline stages.
- Replaced a failure-prone iterative algorithm with a vectorized approach, reducing end-to-end runtime from weeks to hours (~100x speedup).

[Link to Additional Projects](#)

Education

Northeastern University, The Roux Institute, MS in Computer Science 2023

New York City College of Technology, BS in Applied Mathematics 2016

Publications

Published in BMC Research Notes (2023) and Genome Research (2024)

[Link to Full Publications List](#)